

EMIT FIANARANTSOA

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N°MATRICULE : 118125

PARCOURS : DA2I L2

TP LSQL GESTION DE NOTE

Voici un modèle relationnel de la gestion des notes annuelles d'une promotion d'étudiant :

- Etudiant (NEtu, Nom, Prenom)
- Matiere (CodeMat, LibelleMat, CoeffMat)
- Evaluer (#NEtu, #CodeMat, Date, Note)

CREATION DE DATABASE :

```
MariaDB [(none)]> use gestion_note;
Database changed
MariaDB [gestion_note]> show databases;
+-----+
| Database |
+-----+
| air_maroc |
| bdw_tp1 |
| gestion_film |
| gestion_location_vaisselle |
| gestion_note |
| gestion_projet |
| information_schema |
| location_vaisselle |
| mysql |
| performance_schema |
| phpmyadmin |
| test |
| user |
+-----+
13 rows in set (0.001 sec)
```

CREATION DES TABLES :

```
MariaDB [gestion_note]> create table Etudiant(NEtu int auto_increment primary key not null, Nom varchar(30) not null, Prenom varchar(20) not null);
Query OK, 0 rows affected (0.062 sec)

MariaDB [gestion_note]> create table Matiere(CodeMat int auto_increment primary key not null, LibelleMat varchar(20) not null, CoeffMat int(5) not null);
Query OK, 0 rows affected (0.034 sec)

MariaDB [gestion_note]> create table Evaluer(NEtu int not null, CodeMat int not null, Date date not null, Note decimal(4,2) not null, foreign key (NEtu) references Etudiant(NEtu), foreign key (CodeMat) references Matiere(CodeMat));
Query OK, 0 rows affected (0.056 sec)
```

```

MariaDB [gestion_note]> desc Etudiant;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| NEtu  | int(11)       | NO   | PRI | NULL    | auto_increment |
| Nom   | varchar(30)   | NO   |     | NULL    |                |
| Prenom | varchar(20)   | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.014 sec)

MariaDB [gestion_note]> desc Matiere;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| CodeMat    | int(11)       | NO   | PRI | NULL    | auto_increment |
| LibelleMat | varchar(20)   | NO   |     | NULL    |                |
| CoeffMat   | int(5)        | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.015 sec)

MariaDB [gestion_note]> desc Evaluer;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| NEtu  | int(11)       | NO   | MUL | NULL    |       |
| CodeMat | int(11)       | NO   | MUL | NULL    |       |
| Date  | date          | NO   |     | NULL    |       |
| Note  | decimal(4,2)  | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.016 sec)

```

INSERTION DES VALEURS DE CHAQUE TABLE :

INSERTION DES VALEURS DE TABLE ETUDIANT :

```

MariaDB [gestion_note]> insert into Etudiant values(1, 'RAKOTO', 'Andry');
Query OK, 1 row affected (0.004 sec)

MariaDB [gestion_note]> insert into Etudiant values(2, 'RASOANAIVO', 'Fara');
Query OK, 1 row affected (0.008 sec)

MariaDB [gestion_note]> insert into Etudiant values(3, 'RANDRIA', 'Hery');
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Etudiant values(4, 'RAKOTOMALALA', 'Tina');
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Etudiant values(5, 'RAZAFINDRAKOTO', 'Miora');
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Etudiant values(6, 'RAVELOSON', 'Kanto');
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Etudiant values(7, 'RAMANANTSOA', 'Lova');
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Etudiant values(8, 'RABEMANANJARA', 'Tiana');
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Etudiant values(9, 'ANDRIANINA', 'Tahiry');
Query OK, 1 row affected (0.016 sec)

MariaDB [gestion_note]> insert into Etudiant values(10, 'RAMIANDRISOA', 'Fitia');
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> select * from Etudiant;
+-----+-----+-----+
| NEtu | Nom          | Prenom |
+-----+-----+-----+
| 1    | RAKOTO      | Andry  |
| 2    | RASOANAIVO  | Fara   |
| 3    | RANDRIA     | Hery   |
| 4    | RAKOTOMALALA | Tina   |
| 5    | RAZAFINDRAKOTO | Miora |
| 6    | RAVELOSON   | Kanto  |
| 7    | RAMANANTSOA | Lova   |
| 8    | RABEMANANJARA | Tiana |
| 9    | ANDRIANINA  | Tahiry |
| 10   | RAMIANDRISOA | Fitia  |
+-----+-----+-----+

```

INSERTION DES VALEURS DE TABLE MATIERE :

```
MariaDB [gestion_note]> insert into Matiere values(1, 'Algorithmme', 6);
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Matiere values(2, 'Electronique', 2);
Query OK, 1 row affected (0.018 sec)

MariaDB [gestion_note]> insert into Matiere values(3, 'Anglais', 2);
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Matiere values(4, 'Statistique', 3);
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Matiere values(5, 'Reseaux', 3);
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Matiere values(6, 'Systeme', 5);
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Matiere values(7, 'Web', 3);
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Matiere values(8, 'Base de donnees', 4);
Query OK, 1 row affected (0.018 sec)

MariaDB [gestion_note]> insert into Matiere values(9, 'Programmation C', 4);
Query OK, 1 row affected (0.004 sec)

MariaDB [gestion_note]> insert into Matiere values(10, 'Gestion', 3);
Query OK, 1 row affected (0.004 sec)
```

```
MariaDB [gestion_note]> select * from Matiere;
```

CodeMat	LibelleMat	CoeffMat
1	Algorithmme	6
2	Electronique	2
3	Anglais	2
4	Statistique	3
5	Reseaux	3
6	Systeme	5
7	Web	3
8	Base de donnees	4
9	Programmation C	4
10	Gestion	3

INSERTION DES VALEURS DE TABLE EVALUER :

```
MariaDB [gestion_note]> insert into Evaluer values(1, 1, '2025-01-10', 15);
Query OK, 1 row affected (0.004 sec)

MariaDB [gestion_note]> insert into Evaluer values(2, 2, '2025-01-12', 12);
Query OK, 1 row affected (0.004 sec)

MariaDB [gestion_note]> insert into Evaluer values(3, 3, '2025-01-14', 14);
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Evaluer values(4, 4, '2025-01-18', 10);
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> insert into Evaluer values(5, 5, '2025-01-20', 16);
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Evaluer values(6, 6, '2025-01-23', 7);
Query OK, 1 row affected (0.017 sec)

MariaDB [gestion_note]> insert into Evaluer values(7, 7, '2025-01-25', 11);
Query OK, 1 row affected (0.016 sec)

MariaDB [gestion_note]> insert into Evaluer values(8, 8, '2025-01-31', 13);
Query OK, 1 row affected (0.018 sec)

MariaDB [gestion_note]> insert into Evaluer values(9, 9, '2025-02-03', 9);
Query OK, 1 row affected (0.024 sec)

MariaDB [gestion_note]> insert into Evaluer values(10, 10, '2025-02-07', 18);
Query OK, 1 row affected (0.003 sec)

MariaDB [gestion_note]> select * from Evaluer;
+-----+-----+-----+-----+
| NEtu | CodeMat | Date       | Note |
+-----+-----+-----+-----+
| 1    | 1       | 2025-01-10 | 15.00 |
| 2    | 2       | 2025-01-12 | 12.00 |
| 3    | 3       | 2025-01-14 | 14.00 |
| 4    | 4       | 2025-01-18 | 10.00 |
| 5    | 5       | 2025-01-20 | 16.00 |
| 6    | 6       | 2025-01-23 | 7.00  |
| 7    | 7       | 2025-01-25 | 11.00 |
| 8    | 8       | 2025-01-31 | 13.00 |
| 9    | 9       | 2025-02-03 | 9.00  |
| 10   | 10      | 2025-02-07 | 18.00 |
```

1) Le nombre total d'étudiants :

```
SELECT COUNT (*)
FROM Etudiant ;
```

```
MariaDB [gestion_note]> select count(*) from Etudiant;
+-----+
| count(*) |
+-----+
| 10       |
+-----+
1 row in set (0.000 sec)
```

- 2) L'ensemble des notes, la note la plus haute et la note la plus basse :
 SELECT MIN(Note) AS 'plus basse note', MAX(Note) AS 'plus haute note'
 FROM Evaluer ;

```
MariaDB [gestion_note]> select min(Note) as 'plus basse note', max(Note) as 'plus haute
note' from Evaluer;
+-----+-----+
| plus basse note | plus haute note |
+-----+-----+
|          7.00 |          18.00 |
+-----+-----+
1 row in set (0.001 sec)
```

- 3) Les moyennes de chaque étudiant dans chacune des matières :
 SELECT e. NEtu, m. LibelleMat
 AVG (ev. Note) AS MoyEtuMat
 FROM Evaluer ev
 JOIN Matiere m ON ev. CodeMat = m. CodeMat
 JOIN Etudiant e ON e. NEtu =ev. NEtu
 GROUP BY e. NEtu, m. LibelleMat ;

```
MariaDB [gestion_note]> select e.NEtu, m.LibelleMat, AVG(ev.Note) as MoyEtuMat from Evaluer ev join Matiere m ON ev.Code
Mat = m.CodeMat join Etudiant e ON ev.NEtu = e.NEtu group by e.NEtu, m.LibelleMat;
+-----+-----+-----+
| NEtu | LibelleMat | MoyEtuMat |
+-----+-----+-----+
| 1 | Algorithmme | 15.000000 |
| 2 | Electronique | 12.000000 |
| 3 | Anglais | 14.000000 |
| 4 | Statistique | 10.000000 |
| 5 | Reseaux | 16.000000 |
| 6 | Systeme | 7.000000 |
| 7 | Web | 11.000000 |
| 8 | Base de donnees | 13.000000 |
| 9 | Programmation C | 9.000000 |
| 10 | Gestion | 18.000000 |
+-----+-----+-----+
10 rows in set (0.001 sec)
```

- 4) Les moyennes par matière, avec la vue MGETU de la question 3(MOYETUMAT) :
 SELECT LibelleMat, AVG (MoyEtuMat)
 FROM MOYETUMAT
 GROUP BY LibelleMat ;
- 5) SELECT NEtu, SUM(MoyEtuMat*CoeffMat) /SUM (CoeffMat) AS MgEtu
 FROM MOYETUMAT Group By NEtu ;
- 6) La moyenne générale de la promotion, avec la vue MGETU de la question 5 :
 SELECT AVG(MgEtu)
 FROM MGETU ;

- 7) Les étudiants qui ont une moyenne générale supérieure ou égale à la moyenne générale de la promotion, avec la vue MGETU de la question 5 :

```
SEELCT NEtu, Nom, Prenom, MgEtu FROM MGETU WHERE MgEtu >= (SELECT AVG(MgEtu)
FROM MGETU) ;
```

```
MariaDB [gestion_note]> select LibelleMat, AVG(MoyEtuMat) from MoyEtuMat group by Libell
eMat;
ERROR 1146 (42S02): Table 'gestion_note.moyetumat' doesn't exist
MariaDB [gestion_note]> select LibelleMat, AVG(MoyEtuMat) from MOYETUMAT group by Libell
eMat;
ERROR 1146 (42S02): Table 'gestion_note.moyetumat' doesn't exist
MariaDB [gestion_note]> select LibelleMat, AVG(ev.note) from MOYETUMAT group by LibelleM
at;
ERROR 1146 (42S02): Table 'gestion_note.moyetumat' doesn't exist
MariaDB [gestion_note]> select NEtu, SUM(MoyEtuMat*CoeffMat)/SUM(CoeffMat) as MgEtu from
MoyEtuMat group by NEtu;
ERROR 1146 (42S02): Table 'gestion_note.moyetumat' doesn't exist
MariaDB [gestion_note]> select AVG(MgEtu) from MGETU;
ERROR 1146 (42S02): Table 'gestion_note.mgetu' doesn't exist
MariaDB [gestion_note]> select NEtu, Nom, Prenom, MgEtu from MGETU where MgEtu >= (selec
t AVG(MgEtu) from MGETU);
ERROR 1146 (42S02): Table 'gestion_note.mgetu' doesn't exist
```